Indirect Restorations Final Impression Materials and Techniques

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

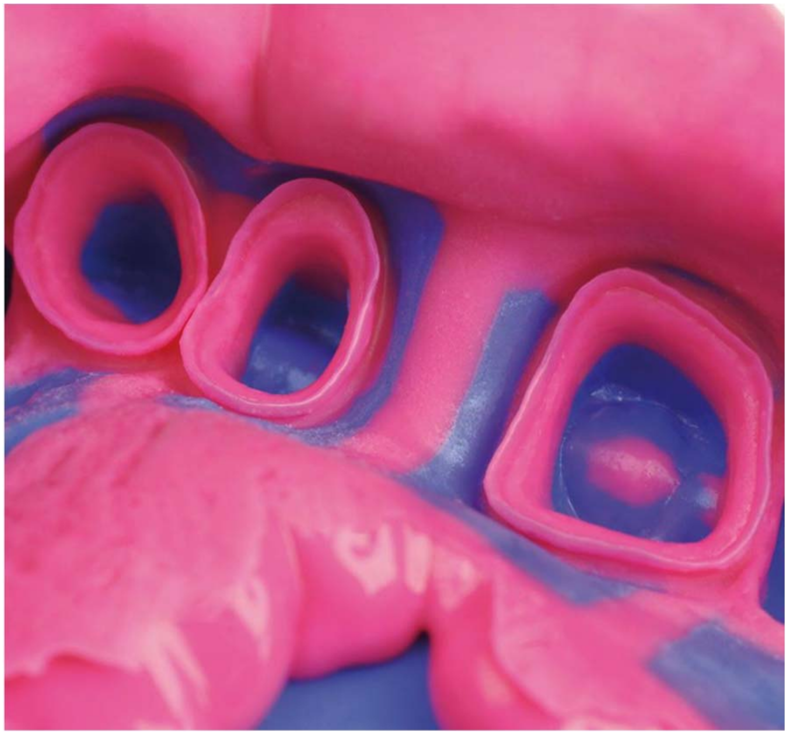
Learning Objectives & Competency Statements

1. Understand the key prerequisites for adequate impressions
2. Know how to make a clinically acceptable impression and how to control for confounding factors
3. Understand the different types of impression materials used to fabricate impressions
4. Develop ability to determine if final impression is clinically acceptable for fabrication of a crown/FDP

Competency statements met #1.4, 3.3, 4.1, 5.1, 5.2

Prerequisites for a good impression

- Adequate **gingival retraction** to capture **prep margins**
- **Tissue management to prevent permanent periodontal damage**
- **Hemostatic agents** to control bleeding as well as **appropriate saliva control**



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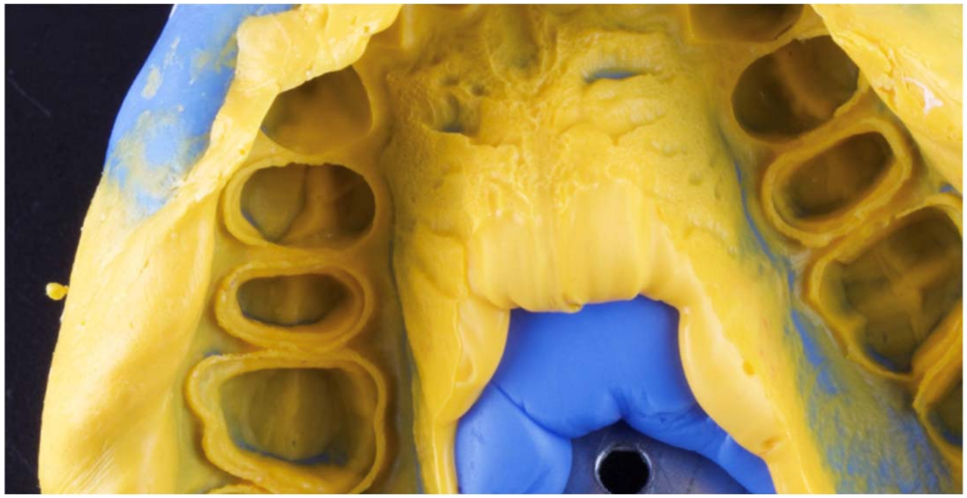
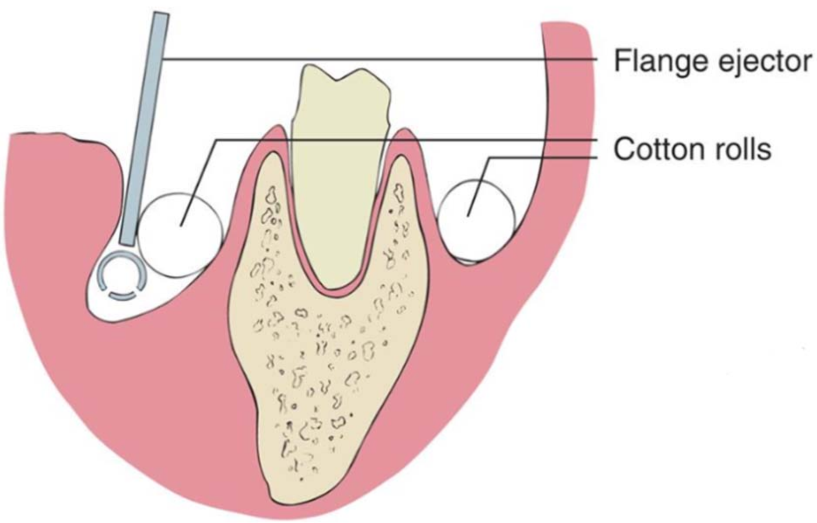
- Appropriate **dimensionally stable** final impression material selected for the purpose
- Impression tray selected and/or modified to optimize quality of the impression
- Possible **laser use** for gingival

troughing?

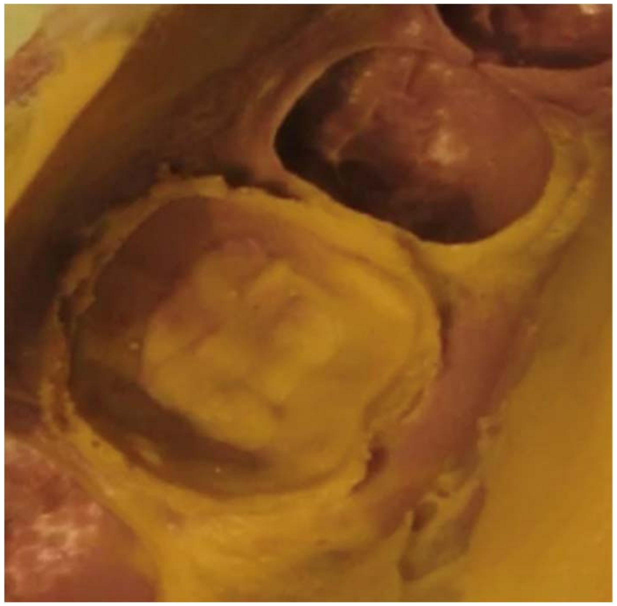


Proper Isolation

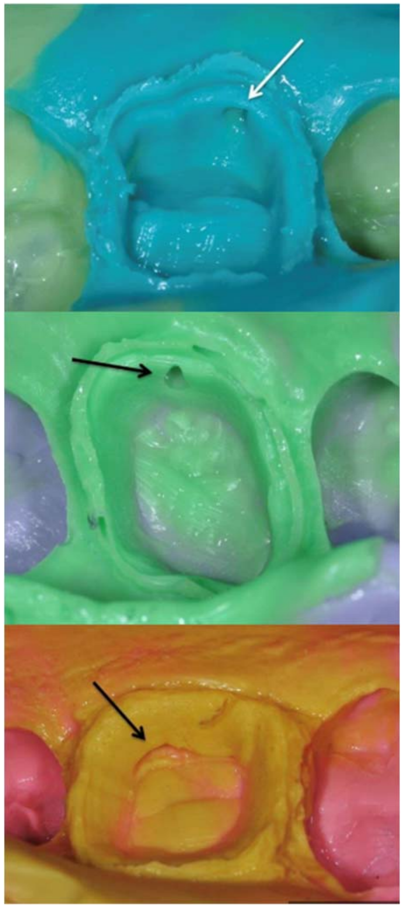
- Most impression materials are **hydrophobic**. What does this mean for you?

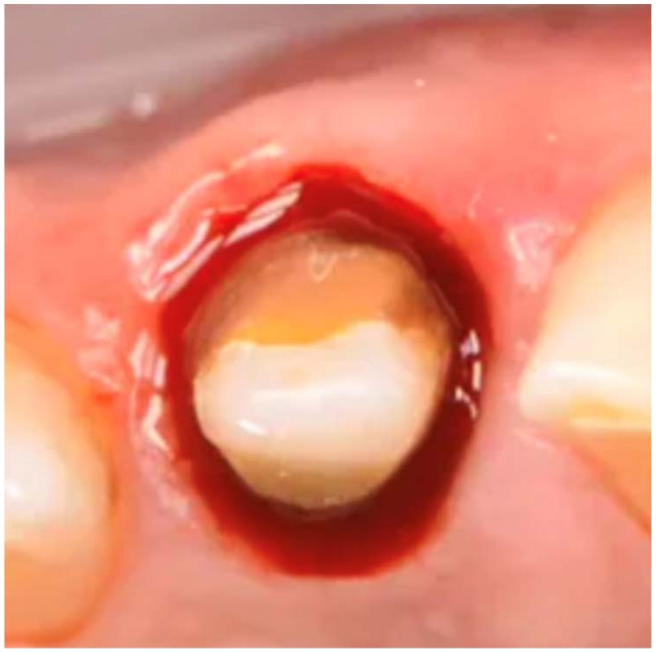


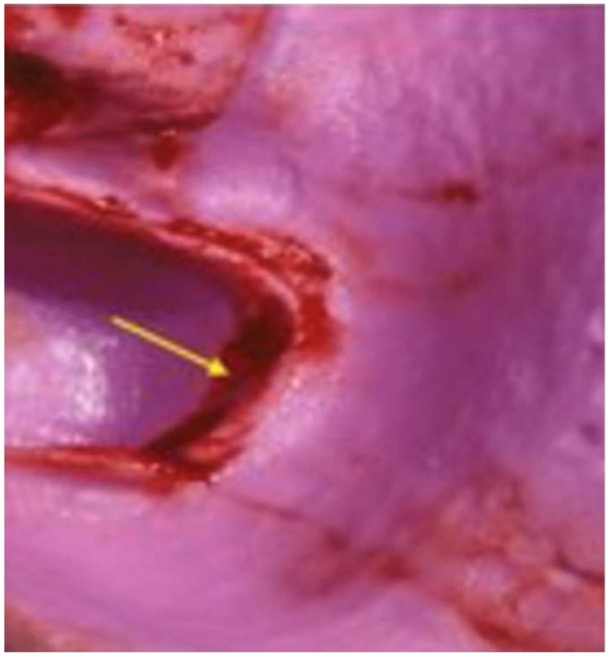
Saliva and Hemostasis



- Improper saliva control will lead to smearing of the material and trapping of air bubble

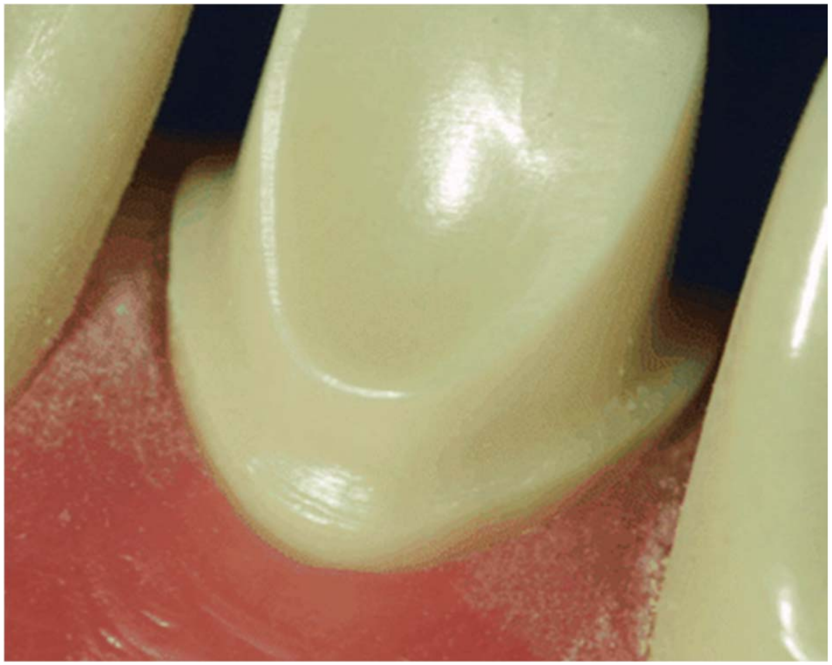






Supragingival Margins

- No need for retraction techniques, however moisture and heme control is still required
- Do not use retraction unless you have to…you could do more damage

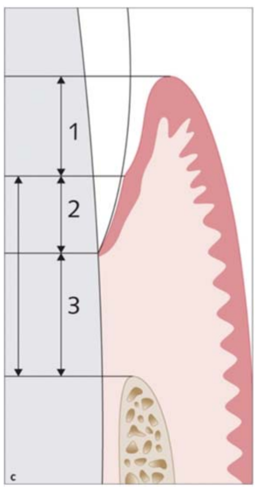


- How do we visualize the margins on a preparation?
- Commonly used are retraction cords and retraction paste.

https://www.styleitaliano.org/step-step-subgingival-preparation-limits/

Cord Retraction

- Single-cord retraction
- Dual-cord retraction

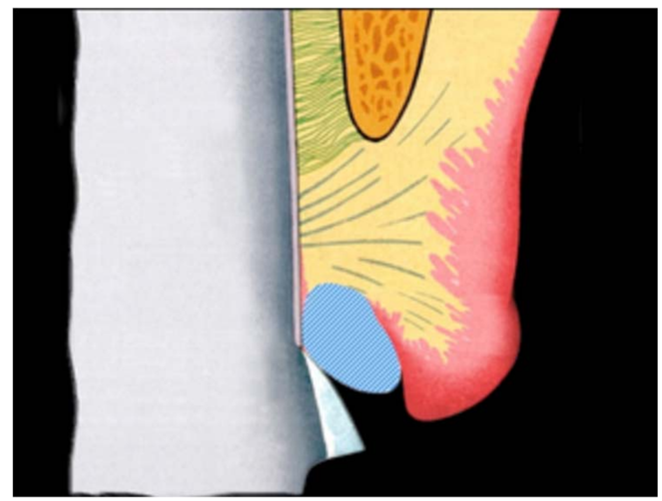


1. Sulcus (~1mm)
2. Junctional epithelium attachment (~1mm)
3. Connective tissue (~1mm)

- If sulcular depth is approx. 1.5mm or less, place subgingival margin approx. 0.5-0.7mm subgingivally.
- If sulcular depths are larger, you can go up to half the depth of the probing depth

https://www.speareducation.com/spear-review/2014/12/biologic-width-part

Single-cord Retraction



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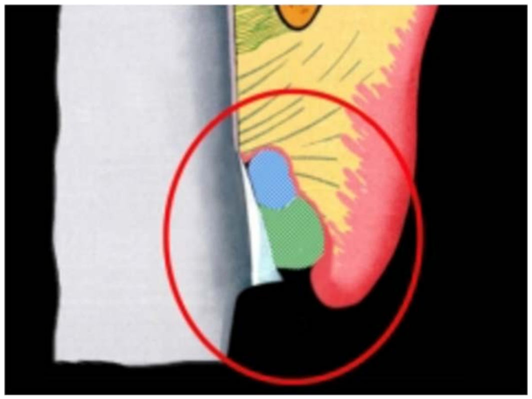
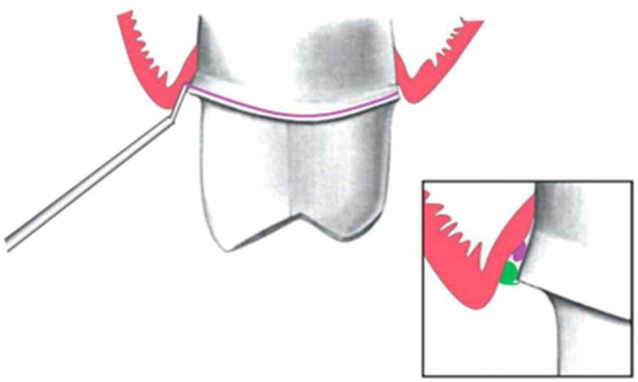
Dual-cord Technique





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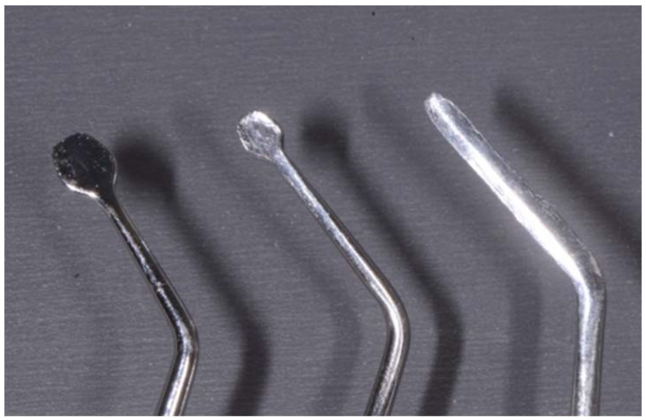
Placing Cord



Cord packers

Choosing the Cord

According to the Ultrapak knitted cords recommendations:

- #000: anterior teeth, use as first cord in dual-cord technique
- #00: prepping and cementing veneers, used in cases of thin gingival phenotype
- #0: lower anteriors, luting veneers, Class III,IV,V restorations, second cord for dual-cord technique
- #1: crown preps
- #2: second cord for dual-cord technique
- #3: very thick tissue phenotype, second cord for dual-cord technique

Source: https://www.ultradent.com/products/categories/retraction/retraction-cord/ultrapak

Gingival Retraction Cords

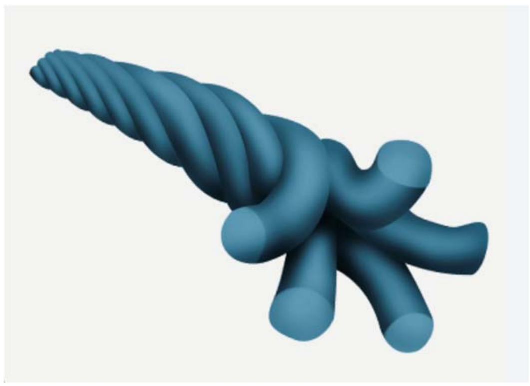
- Many companies, different products. READ the product description and recommendations before ordering!

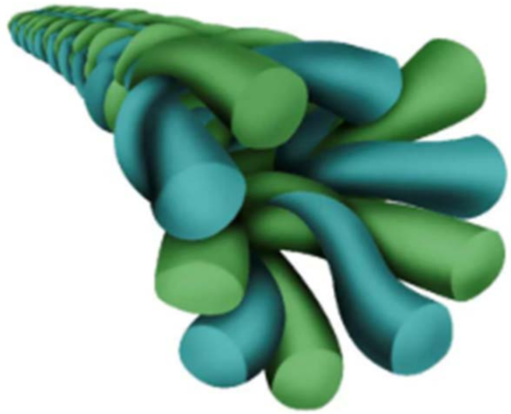
Cord Requirements

- Biocompatible, non-toxic
- Hemostatic, absorbs crevicular fluids
- Ease of insertion and removal
- Color contrast with surrounding tissues

Chauhan R, Chauhan S, Padiyar N, Kaurani P, Gupta A, Khan FN. Present status and future directions: Soft tissue management in prosthodontics. *World J Methodol* 2025; 15(4): 104497 [DOI: 10.5662/wjm.v15.i4.104497]

- Twisted
- Braided
- Knitted





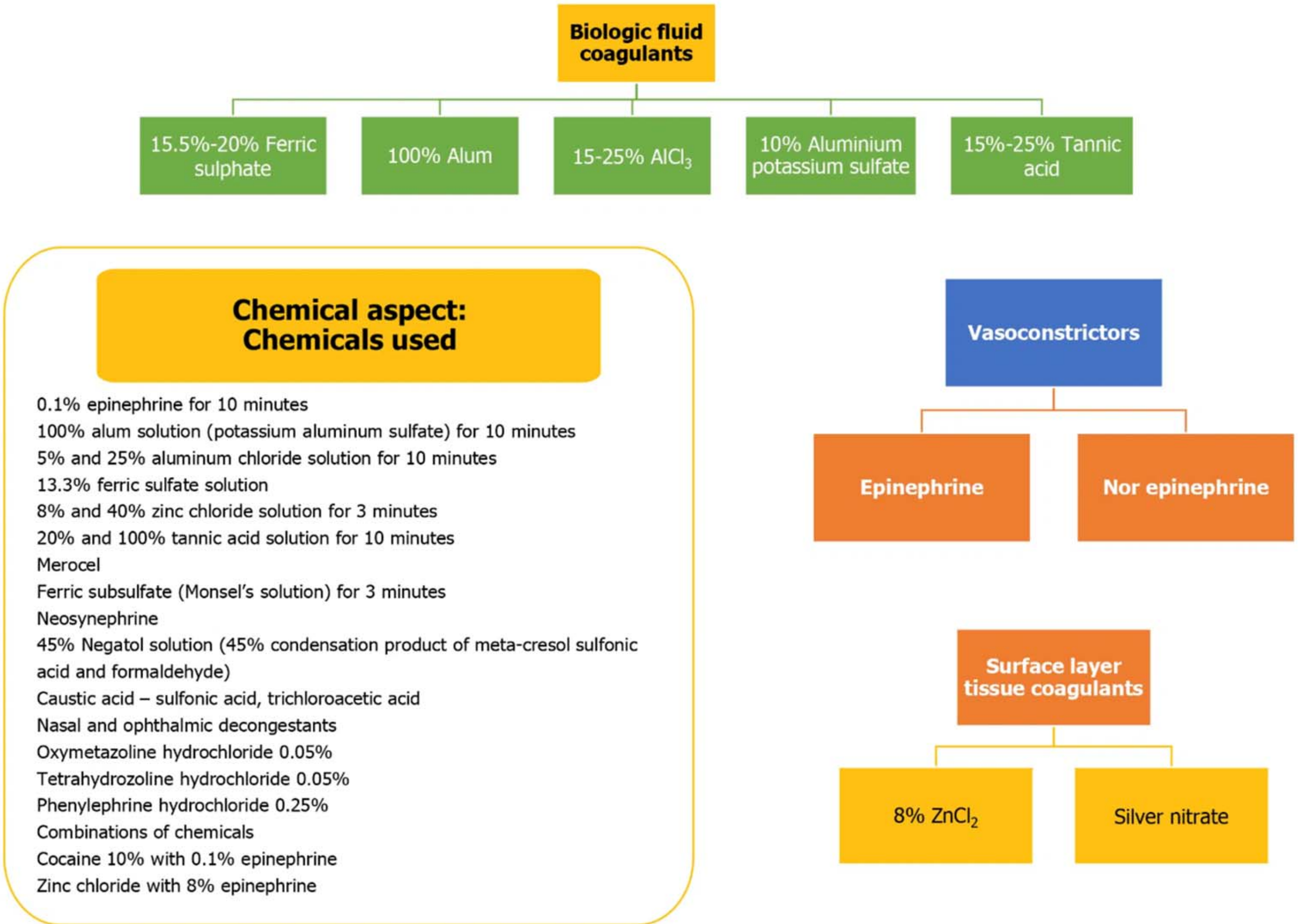


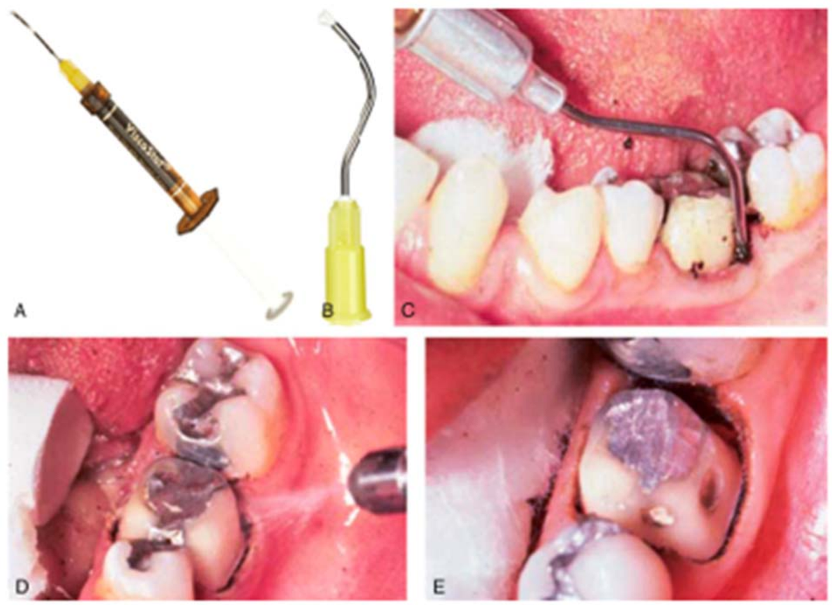
Figure from: Chauhan R, Chauhan S, Padiyar N, Kaurani P, Gupta A, Khan FN. Present status and future directions: Soft tissue management in prosthodontics. *World J Methodol* 2025; 15(4): 104497 [DOI: 10.5662/wjm.v15.i4.104497]

Table 2 The advantages and disadvantages of different chemical agents

Chemicals	Advantages	Disadvantages
Epinephrine	Good tissue displacement	Systemic reactions
	Minimal tissue loss	Epinephrine syndrome
	Good hemostasis	Risk of inflammation of the gingival cuff
		Rebound hyperemia
		Risk of tissue necrosis
Alum	Minimal tissue loss	Less hemostasis and tissue displacement
	Extended working time	Offensive taste
		Risk of necrosis if in high concentration
Aluminum chloride	Minimal tissue loss	Local tissue destruction
	Good hemostasis	Less vasoconstriction than epinephrine
	No systemic effects	Risk of sulcus contamination
	Least irritating of all chemicals	Modifies surface detail reproduction
	Hemostasis	Inhibits set of polyvinyl siloxane and polyether impressions
	Little sulcus collapse after cord removal	
Ferric sulfate	Compatible with aluminum chloride	Non compatible with epinephrine
		Tissue discoloration
	Good displacement	Acidic taste
		Risk of sulcus contamination Inhibits set of polyvinyl siloxane and polyether impressions
Tannic acid	Good tissue response	Less displacement
		Minimal hemostasis

Table from: Chauhan R, Chauhan S, Padiyar N, Kaurani P, Gupta A, Khan FN. Present status and future directions: Soft tissue management in prosthodontics. *World J Methodol* 2025; 15(4): 104497 [DOI: 10.5662/wjm.v15.i4.104497]

- Use of ferric sulfate (ex: ViscoStat, Astringedent) to control bleeding using infuser tip. Use with caution due to reports of cytotoxicity.
- Rinse with water and assess if hemostasis achieved
- Can cause tissue discoloration



Agent	Manufacturer	Active Ingredient	Vehicle	Mean pH
Astringedent	Ultradent	15.5% $Fe_2(SO_4)_3$	Aqueous	0.7
Gingi-Aid	Gingi-Pak	Buffered 25% $AlCl_3$	Aqueous	1.9
Styptin	Van R	20% $AlCl_3$	Glycol	1.3
Hemodent	Premier	21.3% $AlCl_3$-6-hydrate	Glycol (aqueous)	1.2
Hemogin-L	Van R	$AlCl_3$	Aqueous	0.9
Orostat 8%	Gingi-Pak	8% Racemic epinephrine HCl	Aqueous	2.0
ViscoStat	Ultradent	20% $Fe_2(SO_4)_3$	Aqueous	1.6
Aluminum chloride 25%	USP	25% $AlCl_3$	Aqueous	1.1
Stasis	Gingi-Pak	8% Racemic epinephrine HCl	Aqueous	2.0
For comparison: Ketac Conditioner	3M-ESPE Dental	25% Polyacrylic acid	Aqueous	1.7

Table from *Contemporary Fixed Prosthodontics* (6th ed.)

Note the acidity levels of these agents to the tissue.

- Displacement pastes containing $AlCl_3$ (ex: Expasyl) – placed directly in the sulcus, however less gingival retraction can be achieved and careful not to leave in too long – tissue necrosis can occur
- Expanding polymeric foam (ex: Magic FoamCord) with CompreCap to apply pressure

FIG. 14.11 Expanding polymeric foam provides tissue displacement with minimal discomfort or gingival trauma. (A) Magic FoamCord polyvinyl siloxane tissue displacement system. (B) Maxillary incisor prepared for a ceramic crown. Hemorrhage control with ferric sulfate can be used if bleeding is noted (see Fig. 14.9). (C) The expanding polymeric foam is injected around the preparation and condensed with a special hollow cotton roll (ROEKO Comprecap Compression Caps). Expanding foam can also be injected into the hollow of the cotton roll. (D) The patient closes on the cotton roll, maintaining pressure for 5 minutes. (E) Tissue has been displaced from the preparation margins before the impression material is injected or the scan made. (A, Courtesy Coltène, Cuyahoga Falls, Ohio.)

Figures from *Contemporary Fixed Prosthodontics* (6th ed.)

FIG. 14.10 (A) Expasyl is an aluminum chloride–containing paste used for gingival displacement. The material is dispensed from a syringe directly into the sulcus. (B) Fractured ceramic crown had defective margins, which led to significant tissue inflammation and hemorrhage. (C) Crown is removed. (D–F) Paste is directed into the gingival tissues around the prepared margin. (G) After 1 to 2 minutes, the paste is removed with copious amounts of water. (H) Prepared tooth before impression material is injected (I). (A, Courtesy Kerr Corp., Orange, California. B–I, Courtesy Dr. Tony Soileau.)

Electrosurgery

- Contraindications in patients with:
  - Electronic medical devices

    (ex: insulin pumps, TENS devices, pacemakers)
  - Delayed healing
  - Thin gingival biotype

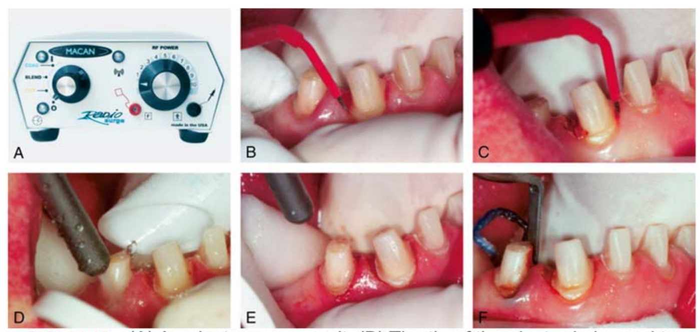
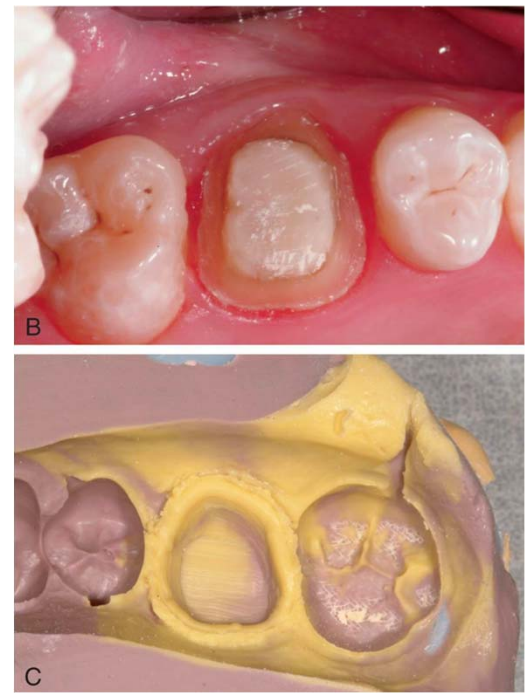


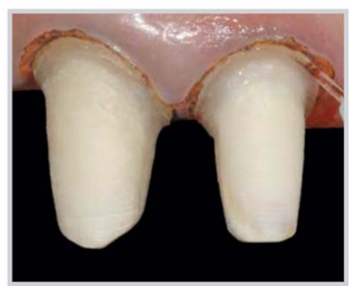
FIG. 14.13 (A) An electrosurgery unit. (B) The tip of the electrode is used to probe the area where the incisions will be made. (C) The tip of the electrode is passed through the hyperplastic tissue. The area is irrigated (D) and dried for inspection (E). (F) After tissue removal, cord placement precedes impression making. (A, Courtesy Macan Engineering Co., Chicago, Illinois.)



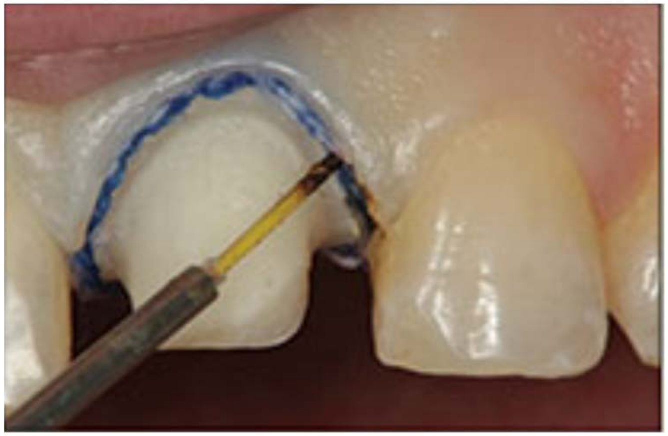
Figures from *Contemporary Fixed Prosthodontics* (6th ed.)

Diode Laser

- Wavelength ranges 810-980nm
- The light is absorbed by melanin and hemoglobin
- Used for crown-lengthening, gingivectomy, frenectomy, removal of hyperplastic tissue, gingival troughing for final impressions



Clinical images: Lee E. A. (2006). Laser-assisted gingival tissue procedures in esthetic dentistry. *Practical procedures & aesthetic dentistry : PPAD*, *18*(9), 2–6.



Images: dentistrytoday.com / Diode Lasers in Restorative Dentistry: An Absolute Necessity for Optimal Care : Dr. Robert A. Lowe

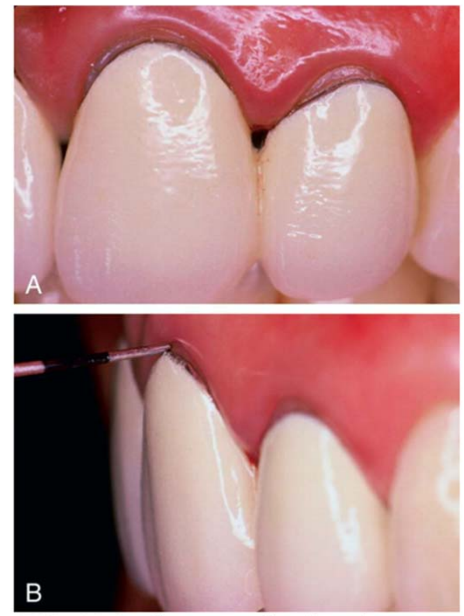


FIG. 14.1 Compromised embrasure form (A) and excessive contours (B) have contributed to the inflammatory response and recession of the gingival tissue.

Images: Shenoy, A., Shenoy, N., & Babannavar, R. (2012). Periodontal considerations determining the design and location of margins in restorative dentistry. *Journal of interdisciplinary dentistry*, *2*(1), 3.

Impression Materials

- Reversible hydrocolloid
- Polysulfide polymer
- Condensation silicone
- Polyether
- Addition silicone

- Irreversible hydrocolloid (ex: alginate) NOT suitable for final impressions

Irreversible Hydrocolloid

- Set time depends on temperature of water
- Pour time should be **less than an hour** after impression, but ideally immediately
- Prone to distortion and needs to be kept in hydrated environment before being poured

Reversible hydrocolloid

- **Syneresis/imbibition**: release and take up of water
- Must be poured shortly after impression is made since it has poor **dimensional stability**
- Custom tray not required, however need special water-cooled trays

FIG. 14.36 Hydrocolloid conditioning equipment consists of three thermostatically controlled water baths: boiling or liquefaction, storage, and tempering. (Courtesy Dux Dental, Oxnard, California.)

FIG. 14.16 Reversible hydrocolloid impression material. Tray (A) and wash material (B). (C) Syringe material. (Courtesy Dux Dental, Oxnard, California.)

Polysulfide polymer

- Should be poured **within an hour** due to lack of dimensional stability
- Use of custom tray to reduce amount of material needed reduces the slight contraction of the material during polymerization
- High tear resistance and overall better dimensional stability than hydrocolloid
- Disadvantages: long intraoral set time of 10 min, odor, need custom trays
- Byproduct of polymerization is **water**

FIG. 14.17 Polysulfide polymers. (Courtesy GC America Inc., Alsip, Illinois.)

Condensation Silicone

- Less dimensional stability than polysulfide
- Short set time of 6-8 min
- Very **hydrophobic** – the area needs to be very dry in order to make a good impression
- Need a surfactant before pouring up impression in stone to avoid voids due to its poor wetting ability. Usually wait 20-30 min before pour
- Byproduct of polymerization is **alcohol**

FIG. 14.18 Condensation silicone. (Courtesy Coltène, Cuyahoga Falls, Ohio.)

Polyether

- Very **good dimensional stability**, can wait to pour impression, but need to store **DRY** because it absorbs water
- Low polymerization shrinkage
- Short set time of 5 min
- Sets via cationic polymerization

(positive ion)

- **NO** polymerization byproducts

FIG. 14.19 Polyether impression material. (Courtesy 3 M ESPE Dental, St. Paul, Minnesota.)

Addition Silicone

- Aka polyvinyl siloxane (PVS)
- Even better dimensional stability than condensation silicone
- Stiffer than polysulfide, less than polyether
- **Hydrophobic**
- **Hydrogen gas** released during polymerization

FIG. 14.20 Addition silicone. (Courtesy GC America Inc, Alsip, Illinois.)

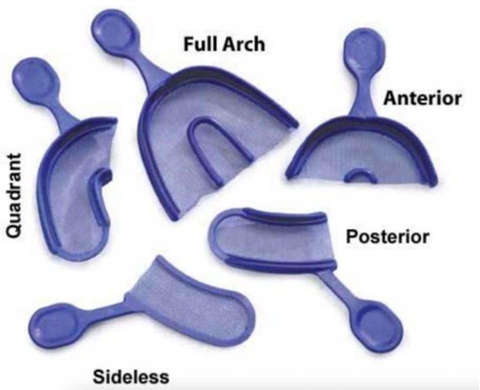
FIG. 14.22 Vinyl polyether silicone. (Courtesy GC America Inc, Alsip, Illinois.)

Light Body and Heavy Body PVS

Custom tray vs. Prefabricated tray

Custom trays can be fabricated from **autopolymerizing acrylic resin, thermoplastic resin, or photopolymerized resin.** They are preferred for increased impression accuracy because they limit amount of material needed to capture desired structures and reduce stress in removal of impression and thermal contraction. You need 2-3mm thickness for appropriate stiffness of tray to avoid distortion of the impression.

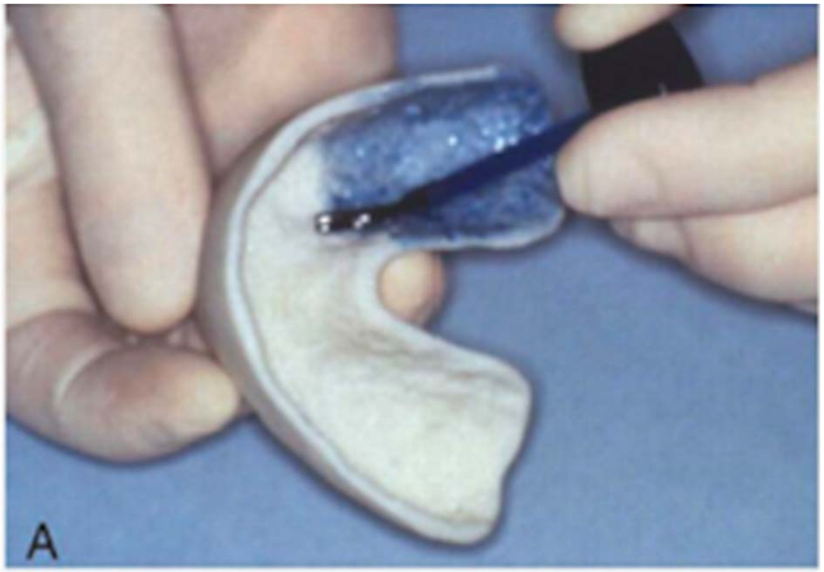
Prefabricated Tray Options

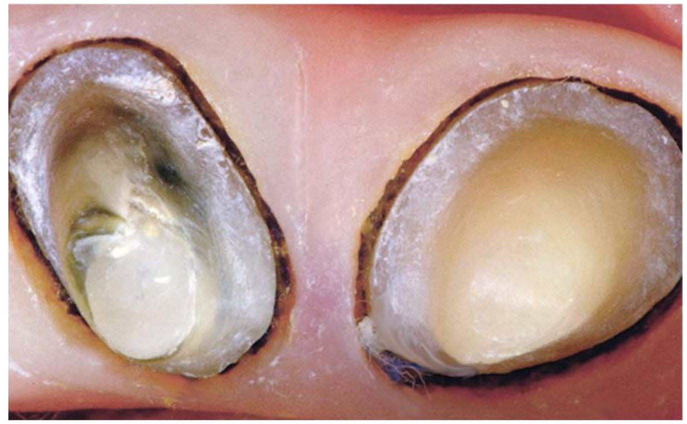


Triple trays

Making a Final Impression

FIG. 14.31 A custom tray should be smooth and well finished. This will enhance its acceptability to the patient.





Images: glidewelldental.com

Summarized Steps

1. Assess your tray – will you use a custom or prefabricated tray? Do adjacent teeth/restorations need to be blocked out with wax so as not to trap material and cause tearing/distortion upon removal.

2. Apply adhesive to the tray.

3. Select cords – Typically #00, and #1, however always let the tissue dictate. Pack cords carefully, leaving a tag on the second cord for easy removal.

4. After about 3-5 min, when removing the top cord, remove in occlusal direction and towards center of tooth, to avoid tissue trauma and localized hemorrhaging which will affect your impression.

5. Lightly dry the abutment teeth. While your assistant is loading the tray with heavy body PVS, you are dispensing light body PVS along the sulcus of the abutment teeth. Dispense in posterior to anterior direction to avoid trapping bubbles. Extend light body material to adjacent teeth occlusal surfaces to capture anatomy for a correct occlusion. ***Make sure to bleed the cartridge a little in the beginning to make sure you have appropriate mixing of the material.

6. Tip: use your air-water syringe to lightly disperse the material, thus ensuring it reaches into the sulcus and captures the margin well.

Continued…

7. Seat in full arch impression tray/triple tray/custom tray and wait the appropriate amount of time until fully set (found in the IFU) Once set, remove impression and assess its accuracy. You should be able to see the margins of the prepped abutment teeth picked up in PVS light body clearly with no bubbles or voids, occlusal surfaces of adjacent teeth without bubble/voids.

- ***Sometimes the second cord will come out with this impression. If that is the case, you can attempt to remove it very carefully so as not to tear the impression, however most of the time, the lab can do that for you.
- Before letting your patient walk out of the office, ALWAYS confirm you have removed ALL cords.

Assess the Impression…

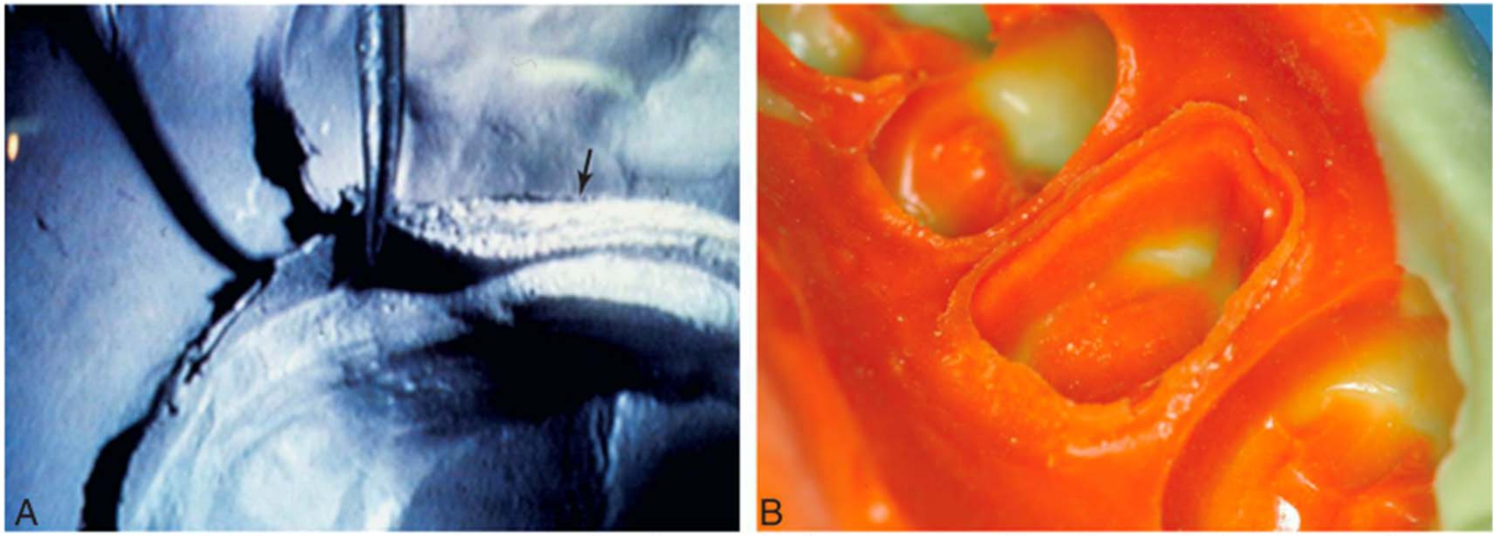


FIG. 14.35 Impression evaluation. (A) Low magnification of elastomeric impression. On the left, an adequate cuff is formed by material extending beyond the preparation margin. On the right side *(arrow)*, the impression does not extend adequately. (B) This impression reproduces an adequate amount of the unprepared tooth structure cervical to the preparation margin.

Occlusal Registration

Images: https://www.dentistrytoday.com/accurate-bite-records-improving-restorative-predictability/

Closed mouth Triple-Tray Technique

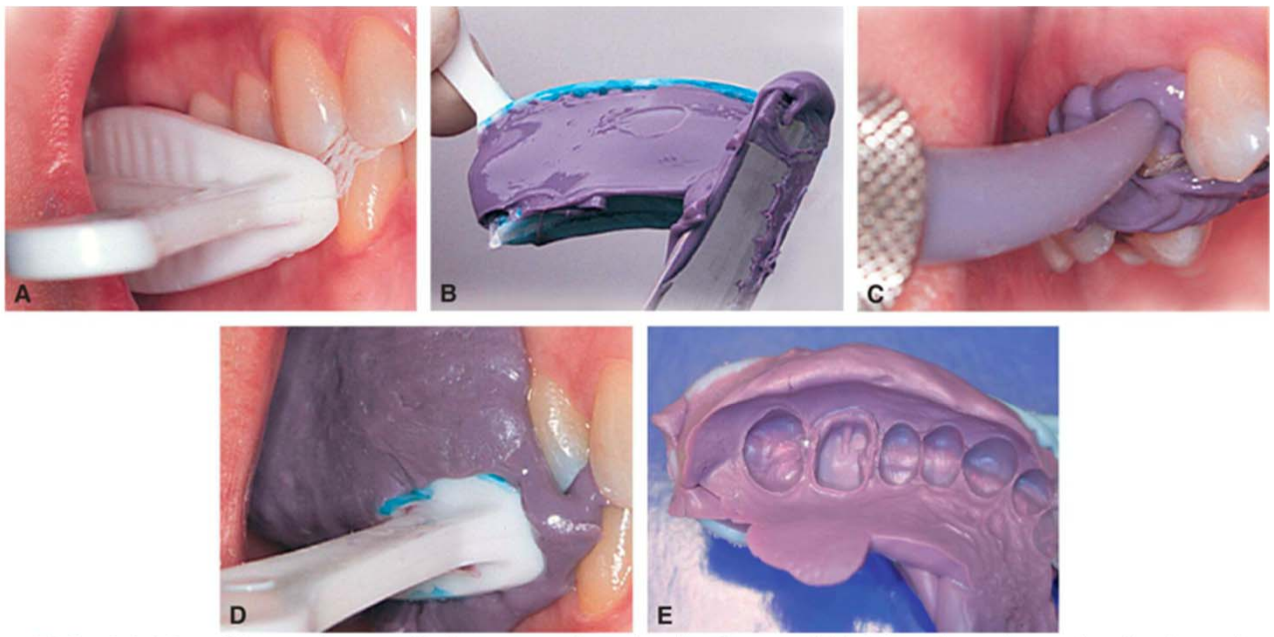


FIG. 14.38 Closed-mouth impression technique. (A) The tray is selected and evaluated. (B) The tray is loaded. (C) Impression material is delivered by syringe. (D) Patient closes into maximum intercuspation. (E) Completed impression. (A–D, Courtesy Premier Dental Products Co., Plymouth Meeting, Pennsylvania.)

Disinfection of Impression

Disinfection	Irreversible[a] Hydrocolloid	Reversible[a] Hydrocolloid	Polysulfide	Silicones	Polyether[b]
Glutaraldehyde 2% (10-min soak time)	Not recommended	Not recommended	Yes	Yes	No
Iodophors (1:213 dilution)	Yes	Yes	Yes	Yes	No
Chlorine compounds (1:10 dilution of commercial bleach)	Yes	Yes	Yes	Yes	Yes
Complex phenolics	Not recommended	Limited data	Yes	Yes	No
Phenolic glutaraldehydes	Not recommended	Yes	Yes	Yes	No

Table from *Contemporary Fixed Prosthodontics* (6th ed.)

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